A boundary case for the rank-four map construction

Human Compression project: L023 working audit

This short, unreviewed calculation concerns Construction 6, Remark 9 and the general class correspondence on page 11 of Leemans–Stokes, arXiv:2208.08215v2. It does not disprove their Class III Theorem 10. The counterexample is a reflexible map with two vertices, two parallel edges and two digonal faces on the sphere; it has eight flags.

Let $G = \mathbb{F}_2^3$ additively, with basis a, b, c, and take the three marked involutions to be a, b, c in that order. A flag of this map can be indexed by the choice of its vertex, edge and face. Changing any one choice gives the corresponding flag adjacency. These three commuting involutions act regularly on the eight flags. In particular this is an actual map with faithful incidence action, as well as a quotient of the source's universal map group. All three periods are two. Every linear automorphism fixing b and permuting the three nonzero vectors in $\langle a, c\rangle$ is a marked Wilson symmetry. There are all six permutations, so the map belongs to Class IV.

Construction 6 gives the four subgroups

$$H_0 = \langle a, b\rangle, \quad H_1 = \langle a, c\rangle, \quad H_2 = \langle b, c\rangle, \quad H_3 = \langle b, a + c\rangle.$$

They are distinct planes in G.

Proposition 1. *The resulting rank-four coset geometry is the complete four-partite incidence structure with two objects of each type. It is firm, and*

$$\mathrm{Aut}(\Gamma) \cong C_2^4, \qquad \mathrm{Cor}(\Gamma) \cong C_2^4 \rtimes S_4.$$

Thus $\mathrm{Cor}(\Gamma)/\mathrm{Aut}(\Gamma) \cong S_4$*, rather than the source's asserted* S_3 *for Class IV.*

Proof. Each subgroup has index two. Distinct planes have sum G, so every coset of one meets every coset of another: for cosets $x + H_i, y + H_j$, express $y - x = u - v$ with $u \in H_i, v \in H_j$ to obtain the common point $x + u = y + v$. Thus all cross-type pairs are incident. Every maximal flag therefore chooses one object from each of the four pairs; there are sixteen chambers. A flag missing one type has exactly two completions, so this geometry is firm.

Independent swaps in the four pairs give every type-preserving permutation, and all preserve incidence. This is C_2^4. Any permutation of the four types, with independent bijections on their pairs, is a correlation, and these exhaust correlations by definition. Their group is $C_2^4 \rtimes S_4$, proving the assertions. □

The face–Petrie residue consists here of two vertices and two edges, with every cross pair incident. It is a four-cycle, not a path on three vertices and two edges. Thus the unrestricted residue description used in source Lemma 8 and Remark 9 fails. The numerical conclusion of Lemma 8 happens to hold for this example: the action of G on chamber labels $(x_0, x_1, x_2, x_3) \in \mathbb{F}_2^4$ has image

$$\{(C, B, A, A + C) : A, B, C \in \mathbb{F}_2\},$$

so has exactly two chamber orbits. No counterexample to that numerical claim is inferred from this calculation.

This is an exact written counterexample to the unrestricted nonfirmness and Class IV correlation-quotient assertions, with no finite computation as premise. Restricting a proposed replacement theorem to another class or to stronger incidence hypotheses would require a separate proof. The general flag-extension claim and the Class III no-duality conclusion remain separate recovery obligations.